



UNIVERSITY OF PETROLEUM AND ENERGY STUDIES
DEHRADUN

DATABASE SYSTEMS LAB

LAB 3

To understand DDL and DML Command

MTECH-COMPUTER SCIENCE ENGINEERING
ARTIFICIAL INTELLIGENCE &
MACHINE LEARNING

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Title: To understand DDL and DML Command

Objective: To understand the concept of designing issues related to the database with creating, populating the tables. To understand the concept of data constraints that is enforced on data being stored in the table. Focus on Primary Key and the Foreign Key.

- Create the tables for Company database as per ER diagram of Exp 2.
- Insert the following data into their respective tables of the Company database.

DDL (Data Definition Language): defines structure of database objects

DDL Commands are blueprint commands. Used to create, change, or remove the structure of the database. Exp: **CREATE, ALTER, DROP, TRUNCATE, RENAME.**

```
6  • ⊕ CREATE TABLE Department (  
14  
15  • ⊕ CREATE TABLE Employee (  
30  
31  • ALTER TABLE Department  
32      ADD CONSTRAINT fk_mgr FOREIGN KEY (Mgr_ssn) REFERENCES Employee(Ssn);  
33  
34  • ⊕ CREATE TABLE Dept_Locations (  
40  
41  • ⊕ CREATE TABLE Project (  
50  
51  • ⊕ CREATE TABLE Works_On (  
59  
60  • ⊕ CREATE TABLE Dependent (  

```

```
--  
6  • ⊕ CREATE TABLE Department (  
7      Dname VARCHAR(15) NOT NULL,  
8      Dnumber INT NOT NULL,  
9      Mgr_ssn CHAR(9),  
10     Mgr_start_date DATE,  
11     PRIMARY KEY (Dnumber),  
12     UNIQUE (Dname)  
13 );  
14  
15  • ⊕ CREATE TABLE Employee (  
16     Fname VARCHAR(15) NOT NULL,  
17     Minit CHAR,  
18     Lname VARCHAR(15) NOT NULL,  
19     Ssn CHAR(9) NOT NULL,  
20     Bdate DATE,  
21     Address VARCHAR(30),  
22     Sex CHAR,  
23     Salary DECIMAL(10,2),  
24     Super_ssn CHAR(9),  
25     Dno INT NOT NULL,  
26     PRIMARY KEY (Ssn),  
27     FOREIGN KEY (Super_ssn) REFERENCES Employee(Ssn),  
28     FOREIGN KEY (Dno) REFERENCES Department(Dnumber)  
29 );  
--  
31  • ALTER TABLE Department  
32      ADD CONSTRAINT fk_mgr FOREIGN KEY (Mgr_ssn) REFERENCES Employee(Ssn);  
33  
34  • ⊕ CREATE TABLE Dept_Locations (  
35     Dnumber INT NOT NULL,  
36     Dlocation VARCHAR(15) NOT NULL,  
37     PRIMARY KEY (Dnumber, Dlocation),  
38     FOREIGN KEY (Dnumber) REFERENCES DEPARTMENT(Dnumber)  
39 );  
40  
41  • ⊕ CREATE TABLE Project (  
42     Pname VARCHAR(15) NOT NULL,  
43     Pnumber INT NOT NULL,  
44     Plocation VARCHAR(15),  
45     Dnum INT NOT NULL,  
46     PRIMARY KEY (Pnumber),  
47     UNIQUE (Pname),  
48     FOREIGN KEY (Dnum) REFERENCES DEPARTMENT(Dnumber)  
49 );  
50  
51  • ⊕ CREATE TABLE Works_On (  
52     Essn CHAR(9) NOT NULL,  
53     Pno INT NOT NULL,  
54     Hours DECIMAL(3,1) NOT NULL,  
55     PRIMARY KEY (Essn, Pno),  
56     FOREIGN KEY (Essn) REFERENCES EMPLOYEE(Ssn),  
57     FOREIGN KEY (Pno) REFERENCES PROJECT(Pnumber)  
58 );  
59  
60  • ⊕ CREATE TABLE Dependent (  
61     Essn CHAR(9) NOT NULL,  
62     Dependent_name VARCHAR(15) NOT NULL,  
63     Sex CHAR,  
64     Bdate DATE,  
65     Relationship VARCHAR(8),  
66     PRIMARY KEY (Essn, Dependent_name),  
67     FOREIGN KEY (Essn) REFERENCES EMPLOYEE(Ssn)  
68 );
```

DML(Data Manipulation Language): manipulates the data inside the tables

DML are the everyday use commands. Used to add, change, delete, or read the actual data inside the database. Exp: **INSERT, UPDATE, DELETE, SELECT.**

```
70 ■ INSERT INTO Department (Dname, Dnumber, Mgr_ssn, Mgr_start_date)
71 VALUES
72 ('Research', 5, NULL, '1988-05-22'),
73 ('Administration', 4, NULL, '1995-01-01'),
74 ('Headquarters', 1, NULL, '1981-06-19');
75
76 ■ INSERT INTO Employee (Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)
77 VALUES
78 ('Franklin', 'T', 'Wong', '333445555', '1955-12-08', '638 Voss, Houston, TX', 'M', 48000, NULL, 5),
79 ('Jennifer', 'S', 'Wallace', '987654321', '1941-06-20', '291 Berry, Bellaire, TX', 'F', 43000, NULL, 4),
80 ('James', 'E', 'Borg', '888665555', '1937-11-10', '450 Stone, Houston, TX', 'M', 55000, NULL, 1);
81
82 ■ UPDATE Department SET Mgr_ssn = '333445555' WHERE Dnumber = 5;
83 ■ UPDATE Department SET Mgr_ssn = '987654321' WHERE Dnumber = 4;
84 ■ UPDATE Department SET Mgr_ssn = '888665555' WHERE Dnumber = 1;
85
86 ■ INSERT INTO Employee (Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)
87 VALUES
88 ('John', 'B', 'Smith', '123456789', '1965-01-09', '731 Fondren, Houston, TX', 'M', 38000, '333445555', 5),
89 ('Alicia', 'J', 'Zelaya', '999887777', '1968-01-19', '3321 Castle, Spring, TX', 'F', 25000, '987654321', 4),
90 ('Ramesh', 'K', 'Narayan', '666884444', '1962-09-15', '975 Fire Oak, Humble, TX', 'M', 30000, '333445555', 5),
91 ('Joyce', 'A', 'English', '453453453', '1972-07-31', '5631 Rice, Houston, TX', 'F', 25000, '333445555', 5),
92 ('Ahmed', 'V', 'Jabbar', '987987987', '1969-03-29', '900 Dallas, Houston, TX', 'M', 25000, '987654321', 4);
93
94 ■ INSERT INTO Dept_Locations (Dnumber, Dlocation)
95 VALUES
96 (1, 'Houston'),
97 (4, 'Stafford'),
98 (5, 'Bellaire'),
99 (5, 'Sugarland'),
100 (5, 'Houston');
101
102 ■ INSERT INTO Project (Pname, Pnumber, Plocation, Dnum)
103 VALUES
104 ('ProductX', 1, 'Bellaire', 5),
105 ('ProductY', 2, 'Sugarland', 5),
106 ('ProductZ', 3, 'Houston', 5),
107 ('Computerization', 10, 'Stafford', 4),
108 ('Reorganization', 20, 'Houston', 1),
109 ('Newbenefits', 30, 'Stafford', 4);
110
111 ■ INSERT INTO Works_On (Essn, Pno, Hours)
112 VALUES
113 ('123456789', 1, 32.5),
114 ('123456789', 2, 7.5),
115 ('666884444', 3, 40.0),
116 ('453453453', 1, 20.0),
117 ('453453453', 2, 20.0),
118 ('333445555', 2, 10.0),
119 ('333445555', 3, 10.0),
120 ('333445555', 10, 10.0),
121 ('333445555', 20, 10.0),
122 ('999887777', 30, 30.0),
123 ('999887777', 10, 10.0),
124 ('987987987', 10, 35.0),
125 ('987987987', 30, 5.0),
126 ('987654321', 30, 20.0),
127 ('888665555', 20, 0);
128
129 ■ INSERT INTO Dependent (Essn, Dependent_name, Sex, Bdate, Relationship)
130 VALUES
131 ('333445555', 'Alice', 'F', '1976-04-05', 'Daughter'),
132 ('333445555', 'Theodore', 'M', '1973-10-25', 'Son'),
133 ('333445555', 'Joy', 'F', '1948-05-03', 'Spouse'),
134 ('987654321', 'Abner', 'M', '1932-02-29', 'Spouse'),
135 ('123456789', 'Michael', 'M', '1988-01-04', 'Son'),
136 ('123456789', 'Alice', 'F', '1988-12-30', 'Daughter'),
137 ('123456789', 'Elizabeth', 'F', '1967-05-05', 'Spouse');
```

1. Created tables using **CREATE DATABASE**;
2. Insert data in the query using **INSERT INTO**;

While creating tables I encountered various errors which are listed at the end of the document.

Table1: Department

Table 2: Employee

Query 1

```

15 CREATE TABLE Employee (
16     Fname VARCHAR(15) NOT NULL,
17     Minit CHAR,
18     Lname VARCHAR(15) NOT NULL,
19     Ssn CHAR(9) NOT NULL,
20     Bdate DATE,
21     Address VARCHAR(30),
22     Sex CHAR,
23     Salary DECIMAL(10,2),
24     Super_ssn CHAR(9),
25     Dno INT NOT NULL,
26     PRIMARY KEY (Ssn),
27     FOREIGN KEY (Super_ssn) REFERENCES Employee(Ssn),
28     FOREIGN KEY (Dno) REFERENCES Department(Dnumber)
29 );
30

```

100% 23:139

Result Grid Filter Rows: Search Edit: Export/Import:

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000.00	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000.00	NULL	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000.00	333445555	5
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000.00	333445555	5
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000.00	NULL	1
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000.00	NULL	4
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000.00	987654321	4
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000.00	987654321	4
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Form Editor Field Types

Table 3: Dept_Locations

Query 1

```
30
31
32 ADD CONSTRAINT fk_mgr FOREIGN KEY (Mgr_ssn) REFERENCES Employee(Ssn);
33
34 CREATE TABLE Dept_Locations (
35     Dnumber INT NOT NULL,
36     Dlocation VARCHAR(15) NOT NULL,
37     PRIMARY KEY (Dnumber, Dlocation),
38     FOREIGN KEY (Dnumber) REFERENCES DEPARTMENT(Dnumber)
39 );
40
41 CREATE TABLE Project (
50
```

100% 30:139

Result Grid Filter Rows: Search Edit: Export/Import:

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Houston
5	Sugarland
NULL	NULL

Result Grid Form Editor

Table 4: Project

Query 1

```
40
41 CREATE TABLE Project (
42     Pname VARCHAR(15) NOT NULL,
43     Pnumber INT NOT NULL,
44     Plocation VARCHAR(15),
45     Dnum INT NOT NULL,
46     PRIMARY KEY (Pnumber),
47     UNIQUE (Pname),
48     FOREIGN KEY (Dnum) REFERENCES DEPARTMENT(Dnumber)
49 );
50
```

100% 23:139

Result Grid Filter Rows: Search Edit: Export/Import:

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4
NULL	NULL	NULL	NULL

Result Grid Form Editor

Table 5: Works_On

Query 1

```
40
41
50
51 CREATE TABLE Works_On (
52     Essn CHAR(9) NOT NULL,
53     Pno INT NOT NULL,
54     Hours DECIMAL(3,1) NOT NULL,
55     PRIMARY KEY (Essn, Pno),
56     FOREIGN KEY (Essn) REFERENCES EMPLOYEE(Ssn),
57     FOREIGN KEY (Pno) REFERENCES PROJECT(Pnumber)
58 );
59
60 CREATE TABLE Dependent (
```

100% 24:139

Result Grid

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
453453453	1	20.0
453453453	2	20.0
666884444	3	40.0
888665555	20	0.0
987654321	30	20.0
987987987	10	35.0
987987987	30	5.0
999887777	10	10.0
999887777	30	30.0
HULL	HULL	HULL

Result Grid
Form Editor
Field Types
Query Stats

Table 6: Dependent

Query 1

```
51 CREATE TABLE Works_On (
59
60 CREATE TABLE Dependent (
61     Essn CHAR(9) NOT NULL,
62     Dependent_name VARCHAR(15) NOT NULL,
63     Sex CHAR,
64     Bdate DATE,
65     Relationship VARCHAR(8),
66     PRIMARY KEY (Essn, Dependent_name),
67     FOREIGN KEY (Essn) REFERENCES EMPLOYEE(Ssn)
68 );
69
```

100% 3:68

Result Grid

Essn	Dependent_name	Sex	Bdate	Relationship
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse
123456789	Michael	M	1988-01-04	Son
333445555	Alice	F	1976-04-05	Daughter
333445555	Joy	F	1948-05-03	Spouse
333445555	Theodore	M	1973-10-25	Son
987654321	Abner	M	1932-02-29	Spouse
HULL	HULL	HULL	HULL	HULL

Result Grid
Form Editor
Field Types

ERRORS ENCOUNTERED / RESOLVED:

1. CREATE TABLE Employee (Fname VARCHAR(15) NOT NULL, Minit CHAR, Lname VARCHAR(15) NOT NULL, Ssn CHAR(9) NOT NULL, Bdate DATE, Address VARCHAR(30), Sex CHAR, Salary DECIMAL(10,2), Super_ssn CHAR(9), Dno INT NOT NULL, PRIMARY KEY (Ssn), FOREIGN KEY (Super_ssn) REFERENCES Employee(Ssn), FOREIGN KEY (Dno) REFERENCES Department(Dnumber))

Error Code: 1824. Failed to open the referenced table 'Department' 0.0056 sec

2. CREATE TABLE Department (Dname VARCHAR(15) NOT NULL, Dnumber INT NOT NULL, Mgr_ssn CHAR(9) NOT NULL, Mgr_start_date DATE, PRIMARY KEY (Dnumber), UNIQUE (Dname), FOREIGN KEY (Mgr_ssn) REFERENCES Employee(Ssn))

Error Code: 1824. Failed to open the referenced table 'Employee' 0.0017 sec

Employee was referenced to Department & Department was Referenced to Employee

Problem Query:

TABLE 1: EMPLOYEE

```
[ Fname VARCHAR(15) NOT NULL, Minit CHAR, Lname VARCHAR(15) NOT NULL, Ssn CHAR(9) NOT NULL, Bdate DATE, Address VARCHAR(30), Sex CHAR, Salary DECIMAL(10,2), Super_ssn CHAR(9), Dno INT NOT NULL, PRIMARY KEY (Ssn), FOREIGN KEY (Super_ssn) REFERENCES EMPLOYEE(Ssn), FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dnumber) ]
```

TABLE 2: DEPARTMENT

```
[Dname VARCHAR(15) NOT NULL, Dnumber INT NOT NULL, Mgr_ssn CHAR(9) NOT NULL, Mgr_start_date DATE, PRIMARY KEY (Dnumber), UNIQUE (Dname), FOREIGN KEY (Mgr_ssn) REFERENCES EMPLOYEE(Ssn) ); ]
```

Resolved Query:

```
6  CREATE TABLE Department (  
7      Dname VARCHAR(15) NOT NULL,  
8      Dnumber INT NOT NULL,  
9      Mgr_ssn CHAR(9),  
10     Mgr_start_date DATE,  
11     PRIMARY KEY (Dnumber),  
12     UNIQUE (Dname)  
13 );  
14  
15 CREATE TABLE Employee (  
16     Fname VARCHAR(15) NOT NULL,  
17     Minit CHAR,  
18     Lname VARCHAR(15) NOT NULL,  
19     Ssn CHAR(9) NOT NULL,  
20     Bdate DATE,  
21     Address VARCHAR(30),  
22     Sex CHAR,  
23     Salary DECIMAL(10,2),  
24     Super_ssn CHAR(9),  
25     Dno INT NOT NULL,  
26     PRIMARY KEY (Ssn),  
27     FOREIGN KEY (Super_ssn) REFERENCES Employee(Ssn),  
28     FOREIGN KEY (Dno) REFERENCES Department(Dnumber)  
29 );  
30  
31 ALTER TABLE Department  
32     ADD CONSTRAINT fk_mgr FOREIGN KEY (Mgr_ssn) REFERENCES Employee(Ssn);  
33
```


**3. INSERT INTO Department (Dname, Dnumber, Mgr_ssn, Mgr_start_date)
VALUES ('Research', 5, '000000000', '1988-05-22'), ('Administration', 4,
'000000000', '1995-01-01'), ('Headquarters', 1, '000000000', '1981-06-19')**

**Error Code: 1452. Cannot add or update a child row: a foreign key
constraint fails ('lab3db`.`department`, CONSTRAINT `fk_mgr` FOREIGN
KEY (`Mgr\_ssn`) REFERENCES `employee` (`Ssn`)) 0.0017 sec**

To resolve this we changed “Mgr_ssn” from NOT NULL to NULL accepted.

```
3  CREATE TABLE Department (  
4      Dname VARCHAR(15) NOT NULL,  
5      Dnumber INT NOT NULL,  
6      Mgr_ssn CHAR(9),  
7      Mgr_start_date DATE,
```

**Due to encountering errors again I took another approach. First I inserted
data to the Department table with Mgr_ssn = NULL. Secondly I inserted the
provided data to the Employee table, after that I updated Mgr_ssn to the
Department table.**

```
59  
60  CREATE TABLE Dependent  
69  
70  INSERT INTO Department (Dname, Dnumber, Mgr_ssn, Mgr_start_date)  
71  VALUES  
72  ('Research', 5, NULL, '1988-05-22'),  
73  ('Administration', 4, NULL, '1995-01-01'),  
74  ('Headquarters', 1, NULL, '1981-06-19');  
75  
76  INSERT INTO Employee (Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)  
77  VALUES  
78  ('Franklin', 'T', 'Wong', '333445555', '1955-12-08', '638 Voss, Houston, TX', 'M', 40000, NULL, 5),  
79  ('Jennifer', 'S', 'Wallace', '987654321', '1941-06-20', '291 Berry, Bellaire, TX', 'F', 43000, NULL, 4),  
80  ('James', 'E', 'Borg', '888665555', '1937-11-10', '450 Stone, Houston, TX', 'M', 55000, NULL, 1);  
81  
82  UPDATE Department SET Mgr_ssn = '333445555' WHERE Dnumber = 5;  
83  UPDATE Department SET Mgr_ssn = '987654321' WHERE Dnumber = 4;  
84  UPDATE Department SET Mgr_ssn = '888665555' WHERE Dnumber = 1;
```

4. INSERT INTO Works_On (Essn, Pno, Hours) VALUES ('123456789', 1, 32.5), ('123456789', 2, 7.5), ('666884444', 3, 40.0), ('453453453', 1, 20.0), ('453453453', 2, 20.0), ('333445555', 2, 10.0), ('333445555', 3, 10.0), ('333445555', 10, 10.0), ('333445555', 20, 10.0), ('999887777', 30, 30.0), ('999887777', 10, 10.0), ('987987987', 10, 35.0), ('987987987', 30, 5.0), ('987654321', 30, 20.0), ('888665555', 20, NULL)

Error Code: 1048. Column 'Hours' cannot be null

0.00086 sec

```
51 CREATE TABLE Works_On (  
52     Essn CHAR(9) NOT NULL,  
53     Pno INT NOT NULL,  
54     Hours DECIMAL(3,1) NOT NULL,  
55     PRIMARY KEY (Essn, Pno),  
56     FOREIGN KEY (Essn) REFERENCES EMPLOYEE(Ssn),  
57     FOREIGN KEY (Pno) REFERENCES PROJECT(Pnumber)  
58 );  
59
```

In the given data one attribute has NULL value for hours. To resolve this we changed the value from NULL to 0.

```
126     ('987654321', 30, 20.0),  
127     ('888665555', 20, 0);
```